

Multiple Choice

1. **Answer:** C
Options: A) 800; B) 830; C) 840; D) 870
Note: Correct option: C - 840
2. **Answer:** C
Options: A) $\frac{\pi}{24}$; B) $\frac{\pi}{12}$; C) $\frac{\pi}{6}$; D) $\frac{\pi}{3}$
Note: Correct option: C - $\frac{\pi}{6}$
3. **Answer:** A
Options: A) $\frac{8}{45}$; B) $\frac{8}{5}$; C) $\frac{8}{7}$; D) $\frac{8}{33}$
Note: Correct option: A - $\frac{8}{45}$
4. **Answer:** D
Options: A) $\frac{7}{2}$; B) $\frac{7}{9}$; C) $\frac{9}{2}$; D) $\frac{9}{7}$
Note: Correct option: D - $\frac{9}{7}$
5. **Answer:** B
Options: A) 6; B) 12; C) 96; D) 102
Note: Correct option: B - 12

True / False

6. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: '-7'.
7. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: '32'.
8. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
9. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: 'the method of selection.'.
10. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** There are six ways in which one European country can be chosen, four in which one Asian country can be chosen, three in which one North American country can be chosen, and seven in which one African country can be chosen. Thus, there are $6 \cdot 4 \cdot 3 \cdot 7 = 504$ ways to form the international commission.

Note: Students should show all steps in their mathematical solution.

12. **Answer:** The distribution of the sample proportion will be less spread out.. Explain why this is the correct answer and provide supporting evidence. Show the calculation or reasoning that leads to this conclusion.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.